



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

October 18, 2023

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 11005
(NINETEENTH REVISION)

EXPIRATION DATE: 2027-09-30

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Missions Systems Orchard Park Inc.
Orchard Park, NY
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of non-DOT specification fiber reinforced plastic (FRP) full composite (FC) aluminum cylinders for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from any Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packagings may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.302(a)(1) and 173.304(a) in that non-DOT specification cylinders are not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Mission Systems Orchard Park Inc. dated August 10, 2023, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed (containing up to 39% by volume oxygen)	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Carbon dioxide	2.2	UN1013	N/A
Helium, compressed	2.2	UN1046	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Methane, compressed <i>or</i> Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Oxygen, compressed	2.2	UN1072	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Prescribed packaging is a non-DOT specification fiber reinforced plastic (FRP) full composite (FC) aluminum cylinder made in conformance with the following:

- (1) Carleton Technologies' Qualification Test Reports, Design/Analysis Reports, and Reports of Design Qualification Testing on file with the Office of Hazardous Materials Safety prior to first shipment; and
- (2) DOT FRP-1 Standard (§ 178.AA) Revision 2 dated February 15, 1987, contained in Appendix A of this special permit, except as follows:

§ 178.AA-2 Type, size and service pressure.

Type 3FC consisting of resin impregnated continuous filament windings in both longitudinal and circumferential directions over a seamless aluminum liner made in compliance with § 178.AA-6; not over 200 pounds water capacity; and service pressure at least 900 psig but not greater than 5,000 psig.

§ 178.AA-15 Marking.

(a) * * *

(b) Required markings are as follows:

(1) "DOT-SP 11005" followed by the service pressure.

(2) A serial number and an identifying symbol (letters); location of number to be just below or immediately following the DOT mark; location of symbol to be just below or immediately following the number. The symbol and numbers must be those of the manufacturer. The symbol must be registered with the Office of Hazardous Materials Safety; duplications not authorized.

* * *

b. TESTING:

(1) The FRP-1 Standard cycling test (§ 178.AA-12(b)) may be conducted at a rate of up to 10 cycles per minute, provided that the minimum dwell time in the pressure range of 90 and 100 percent of the service pressure may not be less than 1.2 seconds.

(2) In lieu of testing for fiber strength per FRP-1 Standard (§ 178.AA-5 (b)) by the cylinder manufacturer, a certificate by the fiber manufacturer is acceptable provided that the cylinder manufacturer's material specification document specifies strength and quality requirements and that the supplied material meets those requirements.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modifications or changes are made to the package and it is offered for transportation in conformance with this special permit and the HMR.

c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

d. Each packaging manufactured under the authority of this special permit must be either (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured or (2) marked with a registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

e. A current copy of this special permit must be maintained at each facility where the packaging is manufactured under this special permit. It must be made available to a DOT representative upon request.

f. Cylinders may not be used for underwater breathing purposes.

g. Cylinders used in oxygen service must conform with §§ 173.302(b)(1) through (b)(4).

h. Cylinder service life may not exceed 15 years from the date of manufacture as marked on the cylinder.

i. Each cylinder must be reinspected and hydrostatically retested every three years in accordance with § 180.205, as prescribed for DOT 3HT specification cylinders, except that the rejection elastic expansion does not apply and the permanent volumetric expansion may not exceed 5 percent of the total volumetric expansion at test pressure. Retest dates must be stamped on the exposed metallic surface of the cylinder neck or marked on a label securely affixed to the cylinder and overcoated with epoxy near the original test date.

Reheat treatment or repair of rejected cylinders is not authorized. When a hydrostatic retest is repeated as provided for in § 180.205(g)(5), only two such retests are permitted.

j. No person may perform inspection and testing of cylinders subject to this special permit unless that person (1) maintains a current copy of this special permit at the location of such inspection and testing, and (2) complies with all the terms and conditions of this special permit.

The marking of the retester's symbol on the cylinders certifies compliance with all of the terms and conditions of this special permit.

k. A cylinder that has been subjected to fire may not be returned to service.

l. Transportation of flammable gases is not authorized aboard cargo vessel or aircraft.

m. Cylinders must be shipped in strong outside packaging in accordance with § 173.301(a)(9).

n. This special permit authorizes the prescribed cylinders only for normal transportation as an article of commerce i.e., the movement of hazardous materials packages from consignor to consignee.

o. Filling requirements are subject to all terms contained in § 173.302a for DOT 3AL specification cylinders.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, cargo-only aircraft, and passenger-carrying aircraft only with passenger carrying aircraft limited to the transportation of compressed air, or nitrogen.

10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel and aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.

11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:

- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
- o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
- o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)—“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

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